

Adam Patterson

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EXPERIENCE

Amazon

Jun. 2024 – Present

Data Scientist (Economist Apprentice)

Seattle, WA

- Developed Surrogate Index Models translating LLM product updates into long-term business impact forecasts
- Reduced production model standard error by 6% via bias-injected simulations
- Built Causal Discovery framework using Peter-Clark (PC) algorithms and DAG inference to identify structural relationships in high-dimensional observational data
- Constructed interactive Agentic Causal Discovery pipeline mapping data inputs to optimal algorithm selection (PC, FCI, GES) via decision rules and LLM reasoning, with human-in-the-loop feedback to enable no-code iteration
- Applied Double Machine Learning (DML) to estimate treatment effects under high-dimensional covariate spaces
- Designed and executed A/B tests to estimate incremental treatment lift and quantify heterogeneous treatment effects, accounting for multiple testing correction, covariate imbalance, and SUTVA violations
- Engineered scalable data pipelines (SQL, Athena, Glue) and automated monthly model runs in AWS via Lambda and SageMaker endpoints; built executive reporting in QuickSight

University of Connecticut

Dec. 2019 – Present

Research Technician

Storrs, CT

- Architected a RAG system for graduate-level econometrics coursework; deployed LLM-based persona agents for survey research
- Managed end-to-end data collection, processing, analysis, and reporting for 6+ published research studies
- Utilized quasi-experimental designs (DiD, event study, IPTW, IV) to quantify causal impacts on student outcomes
- Evaluated heterogeneous treatment effects (HTE) across diverse populations to inform educational policy

Nichols College & Bryant University

Dec. 2022 – Jul. 2024

Adjunct Professor

Dudley, MA & Smithfield, RI

- Taught Data Science (Python, predictive modeling) and Macroeconomics across two institutions
- Supervised undergraduate research team assessing VR intervention impact through pre- and post-surveys

EDUCATION

Central Connecticut State University

2025

M.S. Data Science

University of Connecticut

2019, 2020

B.A. Economics; M.S. Quantitative Economics

SELECTED PUBLICATIONS

- Zhao, W., **Patterson, A.**, et al. (2025). From Metrics to Meaning: Estimating User Feedback Using LLM-Based Evaluation. *Amazon Science*. [\[link\]](#)
 - **Patterson, A.** (2025). Assessing the Alignment of FOMC Statements with Minutes using LLMs. *Issues in Information Systems*, 26(2). [\[link\]](#)
- 10+ peer-reviewed publications, 10+ conference presentations, Distinguished Paper Awards. Full list: [Google Scholar](#)

SKILLS

Technical: SQL, Python, R, AWS (Bedrock, S3, Lambda, SageMaker, Athena, Glue, EC2, ECR, Containers, QuickSight), AutoGluon, PowerBI, Tableau, Excel, LangChain, LangGraph

Statistical: Causal Inference & Discovery, Causal ML, Double Machine Learning, Causal Forests, A/B Testing & Experimentation Design, Treatment Effect Estimation, Regression Analysis, Panel Data, Hypothesis Testing, Machine Learning (Supervised/Unsupervised), Deep Learning, Bayesian Hierarchical Modeling, Time Series Analysis

Domain: Econometrics, Data Science, Agentic AI, Large Language Models, Retrieval Augmented Generation, Natural Language Processing, AutoML, Big Data, Data Visualization